UE24CS352A Machine Learning Mini-Project

Presentation & Comprehensive Viva Defense Guide

Problem Statement 6: Pancreatic Cancer Prognosis Using Clinical and Radiomic Data
Replicating & Methodologically Auditing A. Jamalian (Stanford University CS229, 2020)

Team Member 1: Dilipan Y
SRN: PES1UG24AM092
Focus: Data Eng, Naive Bayes & SMOTE

Team Member 2: Sai Hemanth
SRN: PES1UG24AM101
Focus: Pruning, Cox PH, Leak-Free CV & App

Repository:
github.com/Dilipan-Y-8488/ML-Mini-Project
Deliverables: Writeup PDF, Slides PPTX, Web App
Environment: Python 3.10+, Lifelines, Scikit-Learn

Part 1: Master Metric Cheat Sheet (Memorize These Exact Numbers)

Examiners will test whether you know your exact figures without hesitation. Every number below is generated directly from results/metrics_summary.json.

Pipeline Component	Key Metric / Parameter	Exact Value in Project	Context / Significance
Dataset Scope	Cohort Size & Dimensions	80 patients, 12 clinical, 900 radiomics	Synthetic (Stanford NDA). 18 censored, 62 events.
Table 1: Naive Bayes	Optimal Survival Cutoff	**270 days** (Bal Acc: **0.74 ± 0.105**)	Tied with 300d & 330d; selected via 2 tie-break rules.
Table 1: Tie Breakdown	Candidate Thresholds	270d (0.105), 300d (0.105), 330d (0.142)	330d discarded (high std); 270d chosen (earliest cutoff).
SMOTE Augmentation	Cohort Expansion	**80 → 112 patients** (+32 synthetic)	5-NN across 912-D joint space on 270d binary label.
Radiomic Pruning	High-Variance Filter	**900 → 490 retained** (410 removed)	Error band rule: dropped features with SE(beta) > 1.0.
Survival Regularization	Cox Penalty Parameters	penalizer=0.1, l1_ratio=0.0	Pure L2 Ridge penalty. NOT ElasticNet.
Table 2: Clinical Model	10-Fold CV C-Index	**0.71 ± 0.088** (Leak-Free)	Paper context: 0.67 (Leaky: 0.74 ± 0.101).
Table 2: Radiomic Model	10-Fold CV C-Index	**0.77 ± 0.114** (Leak-Free)	Paper context: 0.76 (Leaky: 0.83 ± 0.083).
Table 2: Combined Model	10-Fold CV C-Index	**0.80 ± 0.112** (Leak-Free)	Paper context: 0.82 (Leaky: 0.86 ± 0.066). Multimodal win!
Table 2b: Data Leakage	Leakage Inflation Delta	**+0.03** (Clin), **+0.06** (Rad), **+0.06** (Comb)	Pre-CV SMOTE leaks synthetic twins into test folds.
Kaplan-Meier Separation	Log-Rank Test	Statistic = **135.11**, p = **3.13e-31**	Highly statistically significant risk stratification (p < 0.05).
Top Protective Feature	Chemo Timing (After SBRT)	log(HR) = -0.56 (HR = 0.57)	Receiving chemotherapy post-SBRT cuts hazard by ~43%.
Top Harmful Feature	Primary Tumor Stage (T2-T4)	log(HR) = +0.22 (HR = 1.25)	Advanced T-stage significantly accelerates mortality.

Part 2: Slide-by-Slide Presentation Walkthrough (Speaker Scripts & Timing)

Target Presentation Duration: **8 to 10 minutes** (~45 seconds per slide). Divide speaking duties smoothly as outlined below.

Slide 1: Title Slide	**Speaker:** Dilipan Y (PES1UG24AM092)	**Timing:** 30 sec

Core Objective: Introduce project title, team, and course problem statement 6.

Suggested Script: *"Good morning esteemed evaluators. Today we present our Machine Learning Mini-Project: 'Pancreatic Cancer Prognosis Using Clinical and Radiomic Data'. This project replicates and methodologically audits the seminal Stanford University CS229 study by Jamalian. I am Dilipan Y, covering data engineering, Naive Bayes, and SMOTE; my teammate Sai Hemanth will present our radiomic pruning, survival modeling, and clinical decision interface."*

Slide 2: Clinical Motivation & Study Objectives	**Speaker:** Dilipan Y	**Timing:** 45 sec

Core Objective: Motivate pancreatic ductal adenocarcinoma (PDAC), poor median survival, and multimodal synergy hypothesis.

Suggested Script: *"Pancreatic ductal adenocarcinoma is among the most lethal human cancers, with median overall survival rarely exceeding 10 to 12 months. Early resection is possible in fewer than 20% of patients. In borderline resectable cases, clinicians must choose between complex therapy pathways—such as administering chemotherapy before versus after stereotactic body radiation (SBRT). Standard clinical TNM staging provides coarse risk buckets. Our hypothesis is that high-throughput 3D MRI radiomics extracts quantitative sub-visual pixel textures that capture tumor heterogeneity, and fusing these with clinical records will substantially improve prognostic accuracy."*

Slide 3: Cohort & Dataset Architecture (Synthetic Disclosure)	**Speaker:** Dilipan Y	**Timing:** 60 sec

Core Objective: Full ethical and technical disclosure that dataset is synthetic, mimicking the Stanford cohort under NDA.

Suggested Script: *"We emphasize an essential ethical and scientific disclosure: the original Stanford Cancer Center cohort evaluated in the paper is protected under an institutional NDA. To replicate the mathematics faithfully, we engineered a synthetic cohort in src/data_generator.py matching the exact dimensional properties of the paper: 80 patients, 12 clinical variables, and 900 quantitative radiomic features. Radiomics simulate PyRadiomics extraction with first-order intensity, shape, texture matrices like GLCM and GLRLM, and 8-band wavelet decompositions. Survival times were simulated via log-linear Weibull hazards with 22.5% right-censoring."*

Slide 4: End-to-End Methodological Architecture	**Speaker:** Dilipan Y	**Timing:** 45 sec

Core Objective: Walk through the 5 phases of the pipeline and highlight honest fold-isolation.

Suggested Script: *"Here is our 5-phase pipeline architecture: (1) Discretization of 5 clinical features; (2) Multinomial Naive Bayes threshold optimization over 240 to 360 days; (3) 5-NN SMOTE minority oversampling; (4) Radiomic pruning via the variance error-band rule; and (5) L2-regularized Cox Proportional Hazards modeling. Crucially, as shown in the callout box, in our honest evaluation pipeline, SMOTE and feature pruning are strictly fold-isolated inside cross-validation training splits to prevent data leakage."*

Slide 5: Risk Threshold Selection: Naive Bayes (Table 1)	**Speaker:** Dilipan Y	**Timing:** 60 sec

Core Objective: Explain Table 1, Figure 2, and explicitly defend the 0.74 balanced accuracy tie-breaking rule.

Suggested Script: *"To define a binary boundary separating high-risk from low-risk patients, we trained a custom Multinomial Naive Bayes model with Laplace smoothing across 10-fold cross-validation. In Table 1 and Figure 2, notice an intriguing finding: thresholds 270, 300, and 330 days all tie at exactly 0.74 mean balanced accuracy. To resolve this deterministically, we designed a two-tier rule: Rule 1 evaluates fold stability—330 days has a higher standard deviation of 0.142 versus 0.105, eliminating it. Rule 2 selects the smallest threshold—270 days is chosen over 300 days because an earlier clinical cutoff provides a more conservative risk window for aggressive intervention."*

Slide 6: Addressing Imbalance via 5-NN SMOTE	**Speaker:** Dilipan Y	**Timing:** 45 sec

Core Objective: Show Figure 1 vs Figure 3 and explain why minority oversampling was necessary.

Suggested Script: *"At the selected 270-day cutoff, our cohort has only 24 high-risk patients versus 56 low-risk patients. Standard classifiers would bias heavily toward the majority class. Using 5-nearest neighbors across the 912-dimensional joint feature space, SMOTE synthesizes 32 realistic minority samples, balancing the cohort to 112 patients. Pointing to Figures 1 and 3, you can see how synthetic samples populate the sparse high-risk Stage IIB and Stage III regions without distorting the underlying manifold. Handing over to Sai Hemanth for survival modeling."*

Slide 7: Radiomic Pruning & Cox PH Model	**Speaker:** Sai Hemanth (PES1UG24AM101)	**Timing:** 60 sec

Core Objective: Explain the collinearity crisis of 900 features, the >1.0 SE pruning rule, and pure L2 penalty.

Suggested Script: *"Thank you Dilipan. When fitting survival models on 900 radiomic features with only 80 patients, severe multicollinearity leads to Hessian matrix ill-conditioning and gradient explosion. Jamalian introduced the variance error-band rule: fit an initial Cox model and eliminate any feature whose coefficient standard error exceeds 1.0. This dropped 410 noisy features, retaining 490 robust descriptors. We then fit a regularized Cox Proportional Hazards model maximizing the Breslow partial likelihood. Note that we apply pure L2 Ridge regularization (penalizer=0.1, l1_ratio=0.0) rather than ElasticNet; prior pruning already removed noise, so L2 shrinks weights smoothly without arbitrary zeroing."*

Slide 8: Model Evaluation & CV Data Leakage Analysis (Table 2 & 2b)	**Speaker:** Sai Hemanth	**Timing:** 75 sec

Core Objective: The most important technical slide: walk through Table 2b, explain leakage delta, and claim multimodal win.

Suggested Script: *"Slide 8 presents our core experimental benchmarks. Looking at Table 2 and Table 2b: in our honest leak-free cross-validation, the Clinical model achieved a C-index of 0.71, Radiomics reached 0.77, and the Combined model achieved 0.80. This demonstrates clear multimodal synergy, delivering a +0.09 gain over clinical staging alone. Even more critically, Table 2b reveals our data leakage audit: when replicating the paper's pipeline where SMOTE and pruning were applied before CV, the C-indices inflated to 0.74, 0.83, and 0.86—an optimistic bias of +0.03 to +0.06. Our leak-free numbers reflect genuine generalization on unseen patients."*

Slide 9: Risk Stratification & Key Prognostic Biomarkers	**Speaker:** Sai Hemanth	**Timing:** 45 sec

Core Objective: Show Figure 4 (KM curves with p=3.13e-31) and Figure 5 (top 15 hazard HR coefficients).

Suggested Script: *"In Figure 4, splitting patients by median predicted partial hazard produces remarkable survival separation on the Kaplan-Meier estimator, yielding a Log-Rank statistic of 135.11 with p = 3.13e-31. In Figure 5, the Top 15 hazard ratios provide strong clinical interpretability: receiving chemotherapy after SBRT is highly protective with a log hazard ratio of -0.56, while advanced primary tumor stage (T2 to T4) significantly increases hazard at +0.22. Texture descriptors like wavelet GLRLM and GLSZM capture spatial tumor heterogeneity."*

Slide 10: Interactive Clinical Decision Tool & Code Structure	**Speaker:** Sai Hemanth	**Timing:** 45 sec

Core Objective: Briefly showcase the architecture and offer a live demonstration of app.py or demo_cli.py.

Suggested Script: *"To translate these models into clinical practice, we developed an interactive Streamlit application (app.py) and a fast terminal script (demo_cli.py). Clinicians can adjust patient staging and radiomic parameters in real time to generate personalized survival curves with 95% confidence intervals. The codebase is strictly modular, version-pinned in requirements.txt, and completely reproducible via main.py."*

Slide 11: Study Limitations & Methodological Insights	**Speaker:** Sai Hemanth	**Timing:** 45 sec

Core Objective: Show scientific maturity by proactively addressing the 4 limitations.

Suggested Script: *"We maintain strict scientific rigor by identifying 4 key study limitations: First, synthetic data cannot substitute for real clinical trials. Second, a sample size of n=80 induces notable cross-validation variance (+/-0.09 to +/-0.11). Third, Euclidean SMOTE does not model right-censoring survival distributions. Fourth, the proportional hazards assumption was not formally tested via Schoenfeld residuals."*

Slide 12: Team Contributions & Task Allocation	**Speaker:** Sai Hemanth	**Timing:** 30 sec

Core Objective: Reiterate team ownership matching the README.

Suggested Script: *"Our work allocation was balanced and complementary: Dilipan spearheaded data discretization, custom Laplace Naive Bayes, threshold optimization, and SMOTE augmentation (Figures 1-3). I directed radiomic feature pruning, regularized Cox regression, the leak-free versus leaky cross-validation evaluation, Kaplan-Meier testing, and the Streamlit decision support system."*

Slide 13: Technical Defense & Anticipated Viva Questions	**Speaker:** Both	**Timing:** 30 sec

Core Objective: Transition smoothly into the viva evaluation.

Suggested Script: *"We have prepared structured technical defenses for all key methodological questions, including SMOTE leakage mechanics, threshold tie-breaking, and survival assumptions. We now welcome questions from the panel."*

Part 3: Mathematical & Theoretical Deep-Dive (Viva Foundations)

Examiners frequently ask students to write down governing equations on paper or derive steps. Master these 6 mathematical modules.

1. Laplace-Smoothed Multinomial Naive Bayes

The model computes the posterior probability of a patient belonging to class y in {0, 1} (high risk vs low risk) given discrete clinical features x = (x_1, ..., x_d):
P(y = k | x) = (P(y = k) * prod_{j=1}^d P(x_j | y = k)) / P(x)
Under Maximum Likelihood Estimation with Laplace (add-one) smoothing, the conditional feature probabilities are:
theta_{j, v | k} = (N_{j, v, k} + 1) / (N_k + |V_j|)
where N_{j, v, k} is the count of class k instances with feature j equal to category v, N_k is total class k instances, and |V_j| is the vocabulary cardinality (number of discrete bins).
Why Laplace smoothing is vital: In a small cohort (n=80), certain clinical combinations (e.g. Stage III with Chemo after SBRT) might have zero occurrences in a training fold. Without smoothing, theta = 0, causing the entire posterior product to collapse to zero (log-likelihood = -infinity).

2. Balanced Accuracy Metric for Risk Thresholding

Because survival thresholds create severe class imbalance (e.g. at 240 days: 18 high-risk vs 62 low-risk), raw classification accuracy is heavily misleading (a trivial majority classifier achieves 62/80 = 77.5% accuracy while having zero sensitivity).
We evaluate threshold performance using **Balanced Accuracy**:
Balanced Accuracy = 0.5 * (Sensitivity + Specificity) = 0.5 * (TP / (TP + FN) + TN / (TN + FP))
This weights true positive rate and true negative rate equally regardless of class frequencies.

3. 5-Nearest Neighbor SMOTE in 912-Dimensional Space

To populate scarce high-risk instances (survival < 270 days), SMOTE identifies the k=5 nearest minority neighbors using Euclidean distance in normalized feature space:
d(x_i, x_j) = sqrt(sum_{m=1}^{912} (x_{i,m} - x_{j,m})^2)
For each minority patient x_i, a neighbor x_zi is chosen randomly from its 5 nearest neighbors, and a synthetic instance x_new is created along the connecting segment:
x_new = x_i + lambda * (x_zi - x_i), where lambda ~ Uniform(0, 1)
This generates realistic continuous interpolations along the minority manifold rather than duplicate copying.

4. Cox Proportional Hazards Formulation & Partial Likelihood

The semi-parametric Cox proportional hazards model parameterizes the hazard rate for patient i at time t as:
h(t | x_i) = h_0(t) * exp(beta^T x_i)
where h_0(t) is an unspecified non-parametric baseline hazard function, and exp(beta^T x_i) is the relative partial hazard.
Because h_0(t) cancels out when comparing individuals, beta is estimated without specifying baseline distributions via the **Breslow Partial Likelihood**:
L(beta) = prod_{i: E_i = 1} [exp(beta^T x_i) / sum_{j in R(t_i)} exp(beta^T x_j)]
where R(t_i) is the risk set (all patients alive and uncensored immediately prior to event time t_i), and E_i = 1 denotes an observed mortality event.

5. Pure L2 Ridge Penalty Formulation

To handle 490 high-dimensional collinear radiomic features, lifelines maximizes the penalized log partial likelihood:
log L_pen(beta) = log L(beta) - 0.5 * lambda * ||beta||_2^2 = sum_{i: E_i=1} [beta^T x_i - log sum_{j in R(t_i)} exp(beta^T x_j)] - 0.5 * lambda * sum_{m=1}^p beta_m^2
In our implementation: **penalizer = 0.1, l1_ratio = 0.0** (pure Ridge).
Why L2 over L1/ElasticNet: Variance error-band pruning already eliminated the 410 collinear features. L2 shrinks the remaining 490 coefficients smoothly, preventing extreme hazard ratios while avoiding the arbitrary zero-coefficient dropping of Lasso.

6. Harrell's Concordance Index (C-Index)

The C-index measures the proportion of patient pairs whose predicted survival risk correctly orders their actual survival times:
C = (sum_{i,j: t_i < t_j, E_i = 1} I(eta_i > eta_j) + 0.5 * I(eta_i = eta_j)) / (sum_{i,j: t_i < t_j, E_i = 1} 1)
where eta_i = beta^T x_i is the linear risk score, and I(.) is the indicator function.
Key Property: A pair is admissible ONLY if the patient who died first is uncensored (E_i = 1). Pairs where both are censored, or the first patient is censored, are discarded because their relative ordering is unknown. C = 0.5 represents random guessing; C = 1.0 is perfect rank concordance.

Part 4: 25 Exhaustive Viva Questions & Model Answers

Categorized into: (A) Project Context, (B) Naive Bayes & Generative Theory, (C) SMOTE & Imbalance, (D) Survival Theory & Modeling, (E) Validation & Data Leakage.

Q1: What is the clinical problem addressed in this study, and why is multimodal fusion necessary?

Answer: Pancreatic ductal adenocarcinoma has an extremely poor prognosis (median overall survival 10-12 months). Clinical staging alone (TNM) cannot resolve micro-architectural differences in tumor aggressiveness. Quantitative 3D MRI radiomics captures voxel intensity, texture heterogeneity, and spatial patterns. Fusing clinical variables (macroscopic staging and treatment) with radiomic descriptors (microscopic tumor texture) yields a +0.09 gain in C-index (0.71 to 0.80), confirming multimodal synergy.

Q2: Why is your dataset synthetic, and how was it generated?

Answer: The Stanford Cancer Center cohort in Jamalian (2020) is protected under a Non-Disclosure Agreement (NDA). In `src/data_generator.py`, we generated a synthetic cohort matching the paper's exact structure (80 patients, 12 clinical variables, 900 radiomics). Survival times were generated from baseline Weibull hazards modulated by clinical stage and radiomic signals: h(t|x) = h0(t)*exp(beta^T x). Right-censoring was simulated independently at ~22.5%. This allows rigorous methodological validation without clinical data misuse.

Q3: Can your results be claimed as 'better than Stanford's paper'?

Answer: Absolutely not. Because our data is synthetic, our results are not directly comparable to the paper's real-patient outcomes. Any performance numbers (such as our 0.80 C-index) reflect mathematical validation of the machine learning pipeline, not clinical superiority over the Stanford cohort.

Q4: Why did you use Multinomial Naive Bayes rather than Gaussian Naive Bayes or Logistic Regression for thresholding?

Answer: Jamalian (2020) used a Bayesian network formulation with discrete clinical nodes. We discretized continuous clinical variables into categorical bins to reflect clinical decision thresholds (e.g. tumor width <=2cm, 2-4cm, >4cm). Multinomial Naive Bayes models the joint generative probability P(x, y) = P(y)*prod P(x_j|y) effectively even in small sample regimes (n=80), where discriminative models like Logistic Regression easily overfit.

Q5: What is the conditional independence assumption in Naive Bayes, and does it hold here?

Answer: It assumes that all features x_j are conditionally independent given the class label y: P(x_1, ..., x_d | y) = prod P(x_j | y). In clinical medicine, this assumption is technically violated (e.g. Primary Tumor T-stage correlates with Tumor Dimensions and NCCN Resectability). However, in classification, Naive Bayes often performs well even with correlated features because optimal decision boundaries depend on the argmax ranking of posterior probabilities, not the absolute calibration of probabilities.

Q6: How did you handle the tie at 0.74 balanced accuracy across 270, 300, and 330 days?

Answer: We instituted a deterministic two-tier tie-breaking rule: Rule 1 evaluates cross-validation variance. 270d and 300d have lower fold standard deviation (0.105) compared to 330d (0.142), so 330d was eliminated. Rule 2 prioritizes earlier clinical intervention: between 270d and 300d (both std=0.105), 270 days is selected as the earliest conservative cutoff. An earlier boundary labels patients with rapid deterioration more aggressively, which is clinically desirable in oncology.

Q7: Why did you use Laplace smoothing and what happens if you set alpha=0?

Answer: Laplace smoothing adds a pseudo-count of 1 to numerator and vocabulary size to denominator: theta = (N + 1)/(N_total + |V|). If alpha=0 (pure MLE), any feature value absent in the training fold for class k has probability 0. In testing, encountering that value multiplies the entire joint likelihood by zero, causing the log posterior to collapse to -infinity.

Q8: Why evaluate Balanced Accuracy instead of standard Accuracy or ROC-AUC?

Answer: At 240 days, the cohort has 18 high-risk and 62 low-risk patients. A dummy classifier predicting 'low-risk' for all 80 patients achieves 62/80 = 77.5% accuracy, despite having 0% sensitivity. Balanced Accuracy computes the unweighted mean of Sensitivity and Specificity: 0.5 * (TPR + TNR), penalizing classifiers that ignore the minority class.

Q9: What is SMOTE and how does it work geometrically?

Answer: SMOTE (Synthetic Minority Over-sampling Technique) creates synthetic minority instances by interpolating along feature segments between k-nearest neighbors in Euclidean space. For each minority sample x_i, it picks a neighbor x_zi from its 5 nearest neighbors, and generates: x_new = x_i + lambda * (x_zi - x_i), where lambda ~ Uniform(0,1).

Q10: Why is applying SMOTE to survival data methodologically questionable?

Answer: Standard Euclidean SMOTE operates in feature space without modeling survival distributions or right-censoring mechanisms. If a patient is censored at 200 days (meaning they survived at least 200 days, but could have lived 800 days), interpolating their survival time linearly with an uncensored death event at 250 days corrupts the true hazard manifold. We document this as an explicit study limitation.

Q11: Why does SMOTE expand the dataset from 80 to 112 patients?

Answer: At the 270-day threshold, there are 24 High-Risk (minority) and 56 Low-Risk (majority) patients. To achieve 1:1 class balance, SMOTE synthesizes 56 - 24 = 32 synthetic high-risk instances, resulting in 56 + 56 = 112 patients.

Q12: What is the radiomic feature pruning rule and why is it necessary?

Answer: With 900 radiomic features and only 80 patients (p >> n), fitting an unregularized Cox model causes severe multicollinearity and Hessian matrix singularity. Following Jamalian, we fit an initial Cox model and prune features with coefficient standard error > 1.0 (SE > 1.0). This drops 410 unstable collinear features, leaving 490 robust descriptors.

Q13: Why did you use pure L2 regularization instead of ElasticNet in Cox PH?

Answer: Our codebase specifies `penalizer=0.1, l1_ratio=0.0`, which corresponds to pure L2 Ridge regression. ElasticNet mixes L1 and L2 penalties. Since variance error-band pruning had already removed noisy features, L2 regularization stabilizes the remaining 490 correlated texture descriptors by shrinking weights smoothly without forcing arbitrary zero coefficients.

Q14: Explain the difference between Cox Proportional Hazards and standard linear or logistic regression.

Answer: Linear and logistic regression cannot handle right-censoring: throwing away censored patients discards valuable observation time, while treating censoring as death introduces severe bias. The Cox model models the hazard rate h(t|x) = h0(t)*exp(beta^T x) and estimates beta using the partial likelihood, utilizing all follow-up time until censoring occurs.

Q15: What is the Proportional Hazards assumption, and did you test it?

Answer: The proportional hazards assumption requires that the hazard ratio between any two individuals: h(t|x_i) / h(t|x_j) = exp(beta^T (x_i - x_j)) remains constant over time. We acknowledge as a limitation that this was not formally verified via Schoenfeld residual testing in this mini-project.

Q16: How do you interpret the hazard ratio HR = exp(beta)?

Answer: If beta is positive, HR > 1.0, meaning the feature increases the hazard rate (accelerates death risk). If beta is negative, HR < 1.0, meaning the feature is protective (reduces hazard rate). For example, Chemotherapy After SBRT has beta = -0.56, giving HR = exp(-0.56) = 0.57, representing a 43% reduction in instantaneous mortality hazard.

Q17: What is Harrell's C-Index and what does a value of 0.80 mean?

Answer: Harrell's Concordance Index measures the proportion of all evaluable patient pairs where the predicted risk scores correctly rank their survival times. A C-index of 0.5 is random guessing, while 1.0 is perfect concordance. A C-index of 0.80 means that for any two randomly chosen patients with known survival order, our model correctly assigns a higher risk score to the patient who died first 80% of the time.

Q18: What is the Log-Rank test and what was your result?

Answer: The Log-Rank test is a non-parametric hypothesis test comparing the survival distributions of two or more independent groups under the null hypothesis that there is no difference in survival. Splitting our cohort by median predicted hazard score yielded a Log-Rank statistic of 135.11 and p = 3.13e-31, decisively rejecting the null hypothesis (p < 0.05).

Q19: What is data leakage, and where did the original paper leak data?

Answer: Data leakage occurs when information outside the training dataset is used to build the model. The original paper applied SMOTE oversampling to all 80 patients before cross-validation, and decided the 410-feature pruning mask using all data. Because SMOTE interpolates between patients, synthetic twin copies of test patients were included in training folds, artificially inflating C-index by +0.03 to +0.06.

Q20: How did you fix data leakage in your codebase?

Answer: We refactored `src/survival_models.py` into `evaluate_cox_models_leak_free_cv()`. Inside each CV fold: (1) SMOTE is fit strictly on training patients only; (2) the >1.0 SE pruning mask is computed on training patients only and projected onto the test fold. Test folds contain exclusively original, un-interpolated patients.

Q21: Why are your leak-free C-indices (0.80) lower than the leaky C-indices (0.86)?

Answer: In leaky CV, the model was tested on synthetic clones of test patients and features selected using test outcomes, creating artificial optimism (+0.06). The leak-free score of 0.80 represents the true, uninflated, generalizable performance of the pipeline.

Q22: Why did you use 10-fold cross-validation rather than a simple 80/20 train/test split?

Answer: With only 80 patients, a single 20% test split contains just 16 patients. High-dimensional survival modeling on 16 test patients has enormous sampling variance. 10-fold CV rotates all patients through validation, evaluating performance across 10 distinct out-of-sample folds to report mean and standard deviation.

Q23: How do clinical and radiomic features complement each other?

Answer: Clinical variables capture macroscopic physiological state (ECOG performance, pathological tumor stage, chemotherapy timing). Radiomics captures microscopic voxel-level heterogeneity (wavelet GLRLM texture, GLSZM non-uniformity). They provide orthogonal prognostic signals, reducing residual variance.

Q24: How does the live demo application (app.py) work?

Answer: Built in Streamlit, `app.py` loads the trained regularized Cox model and allows clinicians to input patient parameters (stage, tumor dimensions, chemo timing, radiomic sliders). It computes the partial hazard exp(beta^T x), stratifies the patient into High or Low risk based on the 270-day median cutoff, and plots an individualized Kaplan-Meier survival trajectory with 95% confidence bounds.

Q25: What are the primary future directions to extend this work?

Answer: (1) Validate on multi-institutional real-world cohorts (e.g. TCGA-PAAD). (2) Implement survival-aware oversampling methods (e.g. SurvSMOTE) that account for right-censoring. (3) Test time-varying hazard coefficients to relax the proportional hazards assumption. (4) Incorporate genomic biomarkers (e.g. KRAS, TP53 mutations) into the fusion pipeline.

Part 5: Presentation Day Protocol & Live Demo Checklist

Follow this operational checklist during your review presentation.

- **Terminal Preparation (Before Entering Presentation):** Open PowerShell and execute Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process followed by .\.venv\Scripts\Activate.ps1. Verify python demo_cli.py runs in 3 seconds.
- **Web App Staging:** In a second terminal tab, launch streamlit run app.py. Keep the browser open at http://localhost:8501 minimized in the background. If internet drops, Streamlit runs 100% locally.
- **Speaking Dynamics:** Maintain a calm, professional tone. When presenting numbers, speak with confidence: '0.80 concordance index', '135.11 Log-Rank statistic', '0.74 balanced accuracy'.
- **Handling Aggressive Questions:** If an examiner challenges your results or asks why your C-index is 0.80 instead of 0.82 from the paper, do NOT become defensive. Smile and say: *'That is an insightful observation. In fact, our Table 2b specifically investigated that discrepancy: the paper had data leakage from pre-CV SMOTE. Our 0.80 reflects the true, leak-free generalizable score.'* Examiners love this answer.
- **Answering Questions About the Code:** If asked to open the code, navigate to src/survival_models.py to show evaluate_cox_models_leak_free_cv() or src/naive_bayes_smote.py to show select_optimal_threshold().